1 (a) $AA^-AA^- = AA^-$ so AA^- is idempotent.

Also, $\text{rank}(AA^-) \le \text{rank}(A) = \text{rank}(AA^-A) \le \text{rank}(AA^-)$

Hence, $I_n - AA^-$ is idempotent $\Rightarrow \text{rank}(I_n - AA^-)$

$= \text{tr}(I_n) - \text{tr}(AA^-)$

$= n - \text{rank}(AA^-)$

$= n - \text{rank}(A)$

$= n - r$

(b) $(I_m - A^-A)^2 = (I_m - A^-A)(I_m - A^-A) = I_m - A^-A - A^-A + A^-A = I_m - A^-A$

so $(I_m - A^-A)$ is idempotent & $\text{rank}(I_m - A^-A) = \text{tr}(I_m - A^-A)$

$= \text{tr}(I_n) - \text{tr}(A^-A)$

$= \text{tr}(I_m) - \text{tr}(AA^-)$

$= m - r$ by above

2 (a) $\text{Var}(\underline{Y}) = \text{Var}(X\underline{b} + \underline{e})$

$= X\,\text{Var}(\underline{b})X' + \text{Var}(\underline{e})$

$= XDX' + \sigma^2 I = V$ (say)

Since $P_X X = X$ & $X'P_X = X'$ (ie $P_X' = P_X$), we have

$P_X V = XDX' + \sigma^2 P_X = V P_X$. Therefore, the SLE is BLUE

(b) Since $\underline{\lambda}$ is estimable, the conjugate NEs have a solution

$(X'X)\underline{r} = \underline{\lambda}$ & the LSE of $\underline{\lambda}'\underline{\beta}$ is $\underline{r}'X'\underline{y}$

with variance $\text{Var}(\underline{r}'X'\underline{Y}) = \underline{r}'X'(X'DX + \sigma^2 I)X\underline{r}$

$= \underline{\lambda}'D\underline{\lambda} + \sigma^2 \underline{r}'X'X\underline{r}$

$(X'X)(X'X)^-X'X$

$= \underline{\lambda}'D\underline{\lambda} + \sigma^2 \underline{\lambda}'(X'X)^-\underline{\lambda}$

3 Here $\underset{n\times p}{X} = [\underset{n\times q}{X_1}\ \underset{n\times(p-q)}{X_2}]$ $\underset{p\times 1}{\underline{\beta}} = \begin{pmatrix} \underline{\beta}_1 \\ \underline{\beta}_2 \end{pmatrix}_{(p-q)\times 1}^{q\times 1}$

& $\underset{(p-q)\times p}{A} = [\underset{(p-q)\times q}{O}\ \underset{p-q}{I}]$ $\underline{b} \in \mathbb{R}^{p-q}$, $A\underline{\beta} = \underline{\beta}_2 = \underline{b}$

$\underline{\lambda}' = (\underline{\lambda}_1', \underline{\lambda}_2')$ is estimable $\Leftrightarrow \underline{\lambda}' \in R\begin{pmatrix} X \\ A \end{pmatrix}$

$\Leftrightarrow (\underline{\lambda}_1', \underline{\lambda}_2') = \underline{g}'\begin{pmatrix} X \\ A \end{pmatrix}$ some $\underline{g} \in \mathbb{R}^{n+p-q}$

$$\Leftrightarrow (\underset{1\times q}{\lambda_1'}, \underset{1\times(p-q)}{\lambda_2'}) = \underset{1\times n}{q'} \begin{bmatrix} X_1 & X_2 \\ 0 & I_{p-q} \end{bmatrix}$$

$$\Leftrightarrow (\underset{1\times q}{\lambda_1'}, \underset{1\times(p-q)}{\lambda_2'}) = \underbrace{(\underset{1\times n}{q_1'}, \underset{1\times(p-q)}{q_2'})}_{q'} \begin{bmatrix} X_1 & X_2 \\ 0 & I_{p-q} \end{bmatrix}$$

$$= (q_1'X_1,\ q_1'X_2 + q_2') \quad \text{holds for some } q_1 \in \mathbb{R}^n,\ q_2 \in \mathbb{R}^{p-q}$$

$\Leftrightarrow \lambda_1' = q_1'X_1$ holds for some $q_1 \in \mathbb{R}^n$ (since we can pick $q_2' = \lambda_2' - q_1'X_2$)

$\Leftrightarrow \lambda_1' \in R(X_1')$

(b) For $\hat{\beta}_1 = (X_1'X_1)^- X_1'(Y - X_2 b)$ & $\hat{\beta}_2 = b$, $\lambda_1'\hat{\beta}_1 + \lambda_2'\hat{\beta}_2$ is the restricted LSE of $\lambda_1'\beta_1 + \lambda_2'\beta_2$ iff $\hat{\beta}_1, \hat{\beta}_2$ are the first part of a solution to the restricted NEs. It suffices to verify that, for some p,

$$\begin{pmatrix} X_1'X_1 & X_1'X_2 & 0 \\ X_2'X_1 & X_2'X_2 & I_{p-q} \\ 0 & I_{p-q} & 0 \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \\ p \end{pmatrix} = \begin{pmatrix} X_1'Y \\ X_2'P_{X_1}(Y - X_2 b) + p \\ b \end{pmatrix} \text{ equals } \begin{pmatrix} X_1'Y \\ X_2'Y \\ b \end{pmatrix}$$

$\left(\begin{pmatrix} X'X & A' \\ A & 0 \end{pmatrix}\right)$

$P_{X_1} = X_1(X_1'X_1)^- X_1'$

$X_1'P_{X_1} = X_1'$

which is true by picking $p = X_2'Y - X_2'P_{X_1}(Y - X_2 b)$. Done.

4(a) the model has $\underset{n\times 3}{X} = [\underset{n\times 1}{1} \ \underset{n\times 1}{x} \ \underset{n\times 1}{z}]$ $\underset{3\times 1}{\beta} = \begin{pmatrix} \mu \\ \alpha \\ \gamma \end{pmatrix}$ $\underset{n\times 1}{\epsilon} \sim$ mean zero, variance σI_n

(b) $X'X = n I_3$, $3 = \text{rank}(I_3) = \text{rank}(X'X) = \text{rank}(X) \le n$

(c) the conjugate equations, for any $\underline{\lambda} = \begin{pmatrix}\lambda_1\\ \lambda_2\\ \lambda_3\end{pmatrix} \in \mathbb{R}^3$,
$X'X\underline{r} = \underline{\lambda}$ have a solution $\underline{r} = \underline{\lambda}/n$
so any $\underline{\lambda}'\underline{\beta}$ is estimable (i.e. all $\underline{\lambda}' \in \mathbb{R}^3$ are in $R(X)$)

(d) Find a solution to NEs as

$$\underbrace{X'X}_{nI_3}\underline{\hat\beta} = X'\underline{Y} = \begin{pmatrix}\Sigma Y_i\\ \Sigma Y_i x_i\\ \Sigma Y_i z_i\end{pmatrix} \Rightarrow \underline{\hat\beta} = \begin{pmatrix}\hat\mu\\ \hat\alpha\\ \hat\gamma\end{pmatrix} = \begin{pmatrix}\frac{1}{n}\sum_{i=1}^n Y_i\\ \frac{1}{n}\sum_{i=1}^n Y_i x_i\\ \frac{1}{n}\sum_{i=1}^n Y_i z_i\end{pmatrix}$$

5 (a) constraint is $r = \mu + \alpha$ or $A\underline{\beta} = 0$
for $A = [1\ 1\ -1]$ & $\underline{\beta} = \begin{pmatrix}\mu\\ \alpha\\ r\end{pmatrix}$, $b = 0$.
any $\underline{\lambda}' \in \mathbb{R}^3 \subset R(X) \subset R\begin{pmatrix}X\\ A\end{pmatrix}$ by 4(c). So any $\underline{\lambda}'\underline{\beta}$ is estimable.

(b) For any $\underline{\lambda} = \begin{pmatrix}\lambda_1\\ \lambda_2\\ \lambda_3\end{pmatrix}$, the constrained conjugate NEs are based on $\begin{pmatrix}X'X & A'\\ A & 0\end{pmatrix} = \begin{pmatrix}n & 0 & 0 & 1\\ 0 & n & 0 & 1\\ 0 & 0 & n & -1\\ 1 & 1 & -1 & 0\end{pmatrix}$ as $\begin{pmatrix}n & 0 & 0 & 1\\ 0 & n & 0 & 1\\ 0 & 0 & n & -1\\ 1 & 1 & -1 & 0\end{pmatrix}\begin{pmatrix}b_1\\ b_2\\ b_3\\ r\end{pmatrix} = \begin{pmatrix}\lambda_1\\ \lambda_2\\ \lambda_3\\ 0\end{pmatrix}$
· for some $b_1, b_2, b_3, r \in \mathbb{R}$

(c) a solution is:
$nb_1 + r = \lambda_1$, $nb_2 + r = \lambda_2$, $nb_3 - r = \lambda_3$, $b_1 + b_2 = b_3$
or $r = \frac{\lambda_1 + \lambda_2 - \lambda_3}{3}$, $b_1 = \frac{1}{n}(\lambda_1 - r) = \frac{1}{n}\left(\frac{2\lambda_1 - \lambda_2 + \lambda_3}{3}\right)$
$b_2 = \frac{1}{n}(\lambda_2 - r) = \frac{1}{n}\left(\frac{2\lambda_2 - \lambda_1 + \lambda_3}{3}\right)$, $b_3 = \frac{1}{n}\left(\frac{2\lambda_3 + \lambda_1 + \lambda_2}{3}\right)$

Under Model IIa, the constrained LSE is BLUE...
this is $(b_1\ b_2\ b_3)\underbrace{X'\underline{Y}}_{\begin{pmatrix}\Sigma Y_i\\ \Sigma Y_i x_i\\ \Sigma Y_i z_i\end{pmatrix}} + r\underbrace{b}_{0}$

$$= \frac{(2\lambda_1 - \lambda_2 + \lambda_3)}{3}\frac{\sum_{i=1}^n Y_i}{n} + \frac{(2\lambda_2 - \lambda_1 + \lambda_3)}{3}\sum_{i=1}^n Y_i x_i/n + \frac{(2\lambda_3 + \lambda_1 + \lambda_2)}{3}\sum_{i=1}^n Y_i z_i/n$$